



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

The Physical Database Structures



Learning Outcome:

Exchange written and spoken information on principles, concepts, data base objects related to query analysis, and data base engines. Describe relational database, its concepts, and principles in spoken or written texts.

Ask your partner:

- Is it easy to organize huge amounts of data in a database, why?
- What can you do to organize the data with ease?

Can Do Statements:

- I can list some components in physical database structures.
- I can talk about characteristics of some components in physical database structures.



Let's Explore!

Mark can't remember the physical database structures' characteristics.



Instructions: Read the text below.

Mark: Remembering all the components of physical structures is hard!

Sam: Let's get through them one by one, so we can keep up with all this information, don't you think so?

Mark: Ok. I recall that datafiles contain all the data. One datafile can only be associated with a tablespace, which helps to organize all the files, and grants people to have easy access to them. When we modify or create new data, it remains in the memory before going to the definitive datafiles; these files are called temporary files or tempfiles.

Sam: In contrast, I know that control files contain the entries that help to get around the physical structure; they control the other type of files to proceed. They have the database name, locations, and timestamps of other files.

Mark: When you say other files, you are talking about the redo logs, right?

Sam: Exactly! You wouldn't happen to know about the redo logs files, would you?

Mark: As a matter of fact, I do. They record all the changes that we made in the data, so if there is a failure or we delete a file, we can back them up. The redo log files are always online, but it's advisable to have archived copies, also called offline copies. I understand all that, but I have trouble grasping the parameter files, don't you have them too?

Sam: No, I don't. They may seem confusing, but they are not. Parameter files simply configure databases, so you can store and manage the data with ease. They usually go hand in hand with the alert and trace log files; when there is that much information, it is really common that errors occur. These files do not fix the errors, they only help you to locate and view them.

Mark: Well, the components aren't hard to recall, but the easiest ones to remember are the backup files.





Let's Practice!

Mark still has doubts about physical database structure.

Instructions: Match column A with column B

COLUMN A

1. We can keep up with all this information,
2. I recall that datafiles contain all the data. One datafile can only be associated with a tablespace,
3. When we modify or create new data, the datafile could take a while to be created,
4. Those are called temporary files, or tempfiles,
5. Control files contains the entries that help to get around the physical structure,

COLUMN B

- a. couldn't it?
- b. don't they?
- c. can't we?
- d. aren't they?
- e. can't they?



Let's Practice!

The Importance of Having Backups!

Instructions: Organize the mystery sentence.

1. there / failure / count / unit. / a / is / we / If / files, / a / with / backup / on / can

2. up / we / I / information. / keep / this / all / think / with / can

3. get / structure. / physical / around / Entries / to / the / help

4. get / component. / each / through / Let's



Extension Activity

More Information about Physical Databases

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about physical database structures.
- Describe each one of them and its usefulness.
- Write a 200 word text explaining each one using phrasal verbs. Don't forget to add examples to support your ideas.
- Discuss your work with your partners.

Self-Assessment

Yes

No

- I deepen my knowledge on physical database structures.
- I express information in a text.
- I support my ideas with my classmates.

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Target Words

- **advisable** (adjective): something good and people recommend it
e.g. It's advisable to have archived copies.
- **ease** (noun): the simplicity of something.
e.g. You can store and manage the data with ease
- **go hand in hand** (verb expression): two people or items that are and work well together.
e.g. They usually go hand in hand with the alert and trace log files
- **take a while** (expression): a short period of time that must pass before something is finished.
e.g. the datafile could take a while to be created
- **get around** (phrasal verb): to move
e.g. Control files contains the entries that help to get around the physical structure
- **as a matter of fact** (expression): in fact
e.g. As a matter of fact, I do



Learning Points

Negative Tag Questions to Suggest and Idea

When you want to be polite or suggest and idea, you can use **negative tag questions**.

Negative questions are a way to suggest and idea or to confirm information. You usually make a statement followed then by the negative question.

Example

We need to buy more soda, **don't we?**

These are the requirements for a negative tag question:

- The statement has to be affirmative:

Example: Carlos **runs** every morning, **doesn't** he?

- It has to keep the tense of the original statement.

Example: Vicky **enjoys** the snow, **doesn't** she?

- The auxiliary verb will depend on the verb used in the statement:

Example: They **will** buy the snacks, **won't** they?

Miguel **can** speak English, **can't** he?

Finally, make sure you're using the correct subject pronoun for your tag question which has to be the subject of the statement.

Vanessa did her homework, didn't **she**?

Examples

- *Let's get through them one by one, so we can keep up with all this information, can't we?*
- *When you say other files, you are talking about the redo logs, aren't you?*





Learning Points

Inseparable Phrasal Verbs

Phrasal verbs are verbs composed of two words that have different meanings when they are separated. There are some phrasal verbs that are separable and there are some others which are inseparable.

Phrasal verbs are verbs composed of two or more words. For example:

Turn: rotate

Up: that points towards or higher

Turn up: increase the volume of a stereo

This phrasal verb can be broken into two without affect its meaning:

*I'll **turn up** the volume.*

*I'll **turn** the volume **up**.*

On the contrary, there are some **phrasal verbs** called "inseparable" because the words can never be set apart.

Examples

(correct) *I **came across** a coin when I went to the bank.*

(incorrect) *I **came** a coin **across** I went to the bank.*

Phrasal Verb	Meaning
back up	support
count on	depend
get rid of	eliminate
keep up with	working to be at the same level than someone
get around	to move around
get through	to overcome a difficult situation.

Examples

- *Let's get through them one by one, so we can keep up with all this information*
- *Control files contain the entries that help to get around the physical structure*



Learning Points

Alert and Trace Log Files

Every procedure in the server including a background process can be copied to an associated trace file. Whenever an error shows up during a process, information about that error is released to its trace file.

It is necessary to know that part of that information is shared with the database administrator as well as the support service. Trace files produce alert logs or files inside databases in a chronological log of messages. The mechanisms that play an important role in the collection of the trace are the Automatic Diagnostic Repository (ADR) and the Incident Packaging Service (IPS). The first one controls and organizes trace files by providing a critical view of critical errors and producing a possible resolution. The second one takes data related to critical errors from ADR and send them to the support services.

